

YOONJI CHOI

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EDUCATION

New York University Courant Institute

Master of Science in Computer Science

Relevant Coursework: Machine Learning, Realtime & Big Data Analysis, Vision Meets ML

New York, New York

Expected May 2027

GPA: 3.45

Stony Brook University

Bachelor of Science in Applied Mathematics and Statistics

Minor in Computer Science | Honors: Cum Laude, Dean's List

Stony Brook, New York

Feb 2022 – May 2025

GPA: 3.63

SKILLS

Programming Languages: Python, Java, R, C, SQL, SAS, MATLAB

ML/AI: Deep Learning, Transformer Models, Statistical Modeling, Interpretability

Tools: PyTorch, scikit-learn, SHAP, pandas, Streamlit

Computing & Systems: High Performance Computing (HPC), OpenMP, MPI, HDFS, Hadoop, Dataproc, Hive, Trino

PROJECTS

Vision Model Training (CNN, ResNet, DETR)

Mar 2026 – May 2026

- Trained and evaluated CNN, ResNet, and Detection Transformer (DETR) models on large-scale image datasets using PyTorch and GPU acceleration.
- Developed end-to-end training pipelines on the NYU Greene HPC cluster using data augmentation, mixed precision, and learning rate scheduling, and analyzed training dynamics under architectural modifications.
- Implemented object matching methods related to DETR and analyzed model behavior across architectures through debugging and performance evaluation.

pico-LLM: Transformer FFN Interpretability Analysis

Nov 2025 – Dec 2025

- Implemented and compared k-gram MLP, LSTM, and Transformer architectures in PyTorch, using Claude Code and GitHub Copilot within a collaborative GitHub workflow.
- Investigated activation patterns in Transformer FFNs, observing progression from lexical to discourse-level representations.
- Designed hypothesis-driven experiments using an activation logging pipeline to examine how Transformer structures influence reasoning behavior and critical evaluation of AI outputs.

Traffic Accidents Causal Analysis

Nov 2025 – Dec 2025

- Built a large-scale ETL pipeline integrating collision, traffic volume, ride-hailing, and weather datasets using distributed data tools including Dataproc, Hive, and Trino.
- Applied Poisson and Negative Binomial regression alongside Random Forest modeling ($R^2 \approx 0.41$) to quantify how environmental and mobility factors influence hourly crash risk across NYC boroughs.
- Identified that snow conditions elevate hourly crash risk by ~25% after controlling for traffic volume and time effects, summarizing findings through model comparisons and visualizations.

Olist Return Analysis

Feb 2025 - May 2025

- Conducted end-to-end analysis on 100K+ e-commerce transactions to examine how customer reviews, purchase behavior, and operational features relate to return patterns under extreme class imbalance (~0.48% return rate).
- Developed predictive modeling pipelines in Python to analyze return-risk patterns, integrating SHAP explanations for model interpretability.
- Deployed a Streamlit dashboard to visualize relationships between customer review patterns, purchasing behavior, and return rates in an interactive format.

RELATED EXPERIENCE

Stony Brook University - Undergraduate Research Assistant

Jan 2025 – May 2025

- Independently developed and optimized parallel PDE/ODE solvers using OpenMP and MPI, improving runtime by up to 1.6x across configurations and delivering performance improvements with minimal day-to-day supervision.
- Validated numerical accuracy against analytical solutions across serial and parallel configurations, maintaining reliability standards in a self-directed research environment.